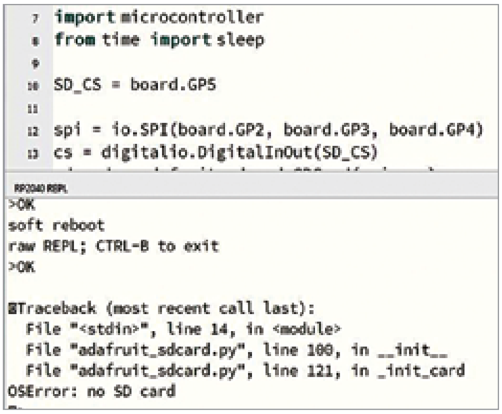
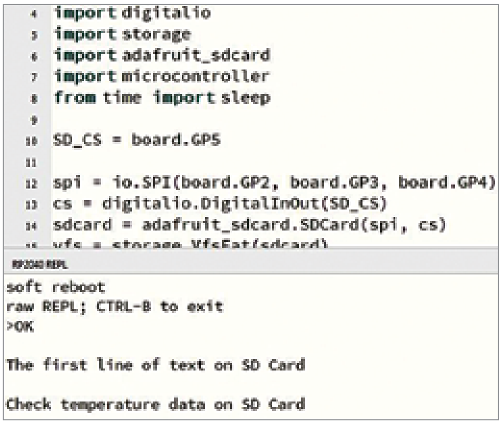


```
import microcontroller
from time import sleep

SD_CS = board.GP5

spi = io.SPI(board.GP2, board.GP3, board.GP4)
cs = digitalio.DigitalInOut(SD_CS)
```

```
RP2040 REPL
>OK
soft reboot
raw REPL; CTRL-B to exit
>OK

Traceback (most recent call last):
  File "<stdin>", line 14, in <module>
  File "adafruit_sdcard.py", line 100, in __init__
  File "adafruit_sdcard.py", line 121, in _init_card
OSError: no SD card
```

Fig. 9: Error message



```
import digitalio
import storage
import adafruit_sdcard
import microcontroller
from time import sleep

SD_CS = board.GP5

spi = io.SPI(board.GP2, board.GP3, board.GP4)
cs = digitalio.DigitalInOut(SD_CS)
sdcard = adafruit_sdcard.SDCard(spi, cs)
vfs = storage.VfsFat(sdcard)
```

```
RP2040 REPL
soft reboot
raw REPL; CTRL-B to exit
>OK

The first line of text on SD Card

Check temperature data on SD Card
```

Fig. 10: Final output

Now, quit Mu editor and unplug the SD card from its adaptor. Insert SD card into the microSD card slot in your laptop/PC and check the various temperature values available in text file format.

Construction and testing

The whole circuit can be assembled on a breadboard or general-purpose PCB. Most Raspberry Pi Pico boards do not come with header pins connected or soldered on the boards. Without these pins, the board cannot be interfaced with external devices, such as sensors, motors, or any device required for the project. Wires can be soldered directly onto the holes provided on the board, but that is a bit risky as you may damage the board.

So, to access GPIO pins, solder each pin carefully using male Berg strip connectors with an appropriate soldering iron. You can then either mount the pins on the breadboard or connect each pin with jumper wire to other components or modules.

After all connections are done, recheck correct polarities, specially Vcc and ground pins. Note that VBUS at pin 40 is typically 5V, which is the main Vcc supply in the circuit. Before inserting the microSD card first format it in FAT32 system and only then insert it into the slot provided in the adaptor.

When you plug in your Raspberry Pi Pico to your laptop the first time, Mu editor attempts to auto-detect your board on start-up and searches for CIRCUITPY drive in the board. So, prior to this, ensure that CircuitPython is properly installed in the drive, as explained above. Next, select the mode and run the code *analog_sensor.py*.

If you get proper sensor output data and waveform, open the main code by clicking on Load option on the main menu bar. Run the code and make sure you get the output message at the bottom of the Mu editor window. You can quit the program and switch off the circuit any time you wish to check the sensor data on the SD card from any computer having microSD card slot. EFY

EFY Note
The source code of this project is available for free download at *source.efymag.com*

Sani Theo is an electronics enthusiast, circuit designer, and technical editor